3.7.1 Inheritance (A-level only)

Content	Opportunities for skills development
The genotype is the genetic constitution of an organism. The phenotype is the expression of this genetic constitution and its interaction with the environment. There may be many alleles of a single gene. Alleles may be dominant, recessive or codominant. In a diploid organism, the alleles at a specific locus may be either homozygous or heterozygous. The use of fully labelled genetic diagrams to interpret, or predict, the results of: • monohybrid and dihybrid crosses involving dominant, recessive and codominant alleles • crosses involving sex-linkage, autosomal linkage, multiple alleles and epistasis. Use of the chi-squared (X^2) test to compare the goodness of fit of observed phenotypic ratios with expected ratios.	**AT h** Students could investigate genetic ratios using crosses of *Drosophila* or Fast Plant® **MS 0.3** Students could use information to represent phenotypic ratios in monohybrid and dihybrid crosses. **MS 1.4** Students could show understanding of the probability associated with inheritance. **MS 1.9** Students could use the X^2 test to investigate the significance of differences between expected and observed phenotypic ratios.